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Simple method for performance evaluation of multistage rockets

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ABSTRACT

Multistage rockets are commonly employed to place spacecraft and satellites in their operational orbits. Performance evaluation of multistage rockets is aimed at defining the maximum payload mass at orbit injection, for specified structural, propulsive, and aerodynamic data of the launch vehicle. This work proposes a simple method for a fast performance evaluation of multistage rockets. The technique at hand is based on three steps: (i) the flight-path angle at each stage separation is guessed, (ii) the spacecraft velocity is maximized at the first and second stage separation, and (iii) for the last stage the thrust direction is obtained through the particle swarm optimization technique, in conjunction with the use of the Euler–Lagrange equations and the Pontryagin minimum principle. The coast duration at the second stage separation is optimized as well. The method at hand is extremely simple and easy-to-implement, but nevertheless it proves to be capable of yielding near-optimal ascending trajectories for a multistage launch vehicle with realistic structural, propulsive, and aerodynamic characteristics. The solutions found with the technique under consideration can be employed either for a rapid evaluation of the multistage rocket performance or as guesses for more refined optimization algorithms.

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1. Introduction

Multistage rockets are commonly employed to place spacecraft and satellites in their operational orbits. Performance evaluation of multistage rockets is basically aimed at defining the maximum payload mass that can be inserted in the desired orbit. Usually, if the rocket characteristics are specified, this evaluation is obtained from optimizing the rocket trajectory, i.e., by determining the optimal control law that leads to maximizing the final mass at orbit injection.

In general, the numerical solution of aerospace trajectory optimization problems is not trivial and has been pursued with different approaches in the past. Indirect methods, such as the gradient-restoration algorithm [1,2] and the shooting method [3], or direct techniques, such as direct collocation [4,5], direct transcription [6,7], and differential inclusion [8,9], are to name a few. However, only a relatively small number of publications are concerned with trajectory optimization of multistage launch vehicles [1,2,10–17,19,20]. Calise et al. [10] and Gath and Calise [11] proposed and applied a hybrid analytic/numerical approach, based on homotopy and starting with the generation of the optimal solution in a vacuum. They adopted the approximate linear gravity model, and the same did Lu and Pan [12] and Lu et al. [13], who applied a multiple-shooting method to optimizing exoatmospheric trajectories composed of two powered phases separated by a coast arc. Weigel and Well [14] used a similar indirect, multiple-shooting approach to analyze and optimize the ascent trajectories of two launch

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vehicles with splash-down constraints. Miele [2] developed and applied the indirect multiple-subarc gradient restoration algorithm to optimizing the two-dimensional ascending trajectory of a three-stage rocket in the presence of dynamical and control constraints. The previously cited works [2,10–14] make use of indirect algorithms and require a considerable deal of effort for deriving the analytical conditions for optimality and for the subsequent programming and debugging. Furthermore, these methods can suffer from a slow rate of convergence and an excessive dependence on the starting guess. This difficulty has been occasionally circumvented through homotopy [10,11,15], but this adds further complexity to the solution process. Other papers deal with direct numerical techniques applied to multistage rocket trajectory optimization. Roh and Kim [16] used a collocation method for optimizing the performance of a four-stage rocket, whose two-dimensional trajectory was assumed to be composed of three thrust phases and a coast arc of specified duration. Collocation was also employed by Jamilnia and Naghash [17], with the additional task of determining the optimal staging, and by Martinon et al. [15], for the purpose of validating the numerical results attained through indirect shooting. This latter paper refers to the Ariane V launch vehicle and is specifically devoted to investigating singular arcs. Direct methods convert the optimal control problem into a nonlinear programming problem involving a large number of unknown parameters to optimize. The disadvantage is in the need of using specialized nlp solvers, such as SNOPT [18]. Recently, different methodologies appeared that do not belong to the category of neither the indirect nor the direct techniques. These techniques are usually referred to as heuristics and have been sporadically applied to optimizing ascent trajectories of multistage rockets. Bayley et al. [19] used a genetic algorithm for the purpose of minimizing the overall rocket mass in the context of a high fidelity model of the entire vehicle. Three- and four-stage rockets were considered, with ascending trajectories composed (respectively) of three or four powered arcs. Lastly, Qazi et al. [20] integrated neural networks, particle swarm optimization, and sequential quadratic programming for the simultaneous conceptual design and trajectory optimization of a new multistage launch vehicle.

The work that follows is concerned with a novel approach, which is intended to supply a fast performance evaluation for multistage rockets with given characteristics, under some simplifying assumptions. The technique described in this work is applied to a three-stage rocket, whose three-dimensional trajectory is composed of the following thrust phases and coast arcs:

(a) first stage propulsion
(b) second stage propulsion
(c) coast arc (after the second stage separation)
(d) third stage thrust phase.

In general, the inclusion of a coast arc (between two powered phases) leads to substantial propellant savings and this circumstance justifies the trajectory structure assumed in this research. Usually the coast duration increases as the injection altitude increases, as remarked by Lu et al. [13].

Specifically, the methodology presented and applied in this paper is based on three steps:

(1) the flight-path angle at each stage separation is determined through an iterative process;
(2) for each stage, the native MATLAB routine *fmincon* is employed for finding the thrust direction that maximizes the velocity at the stage separation, at which the flight-path angle is constrained to the value found at step (1);
(3) for the third stage, the existence and duration of a coast arc and the optimal thrust direction are determined through a heuristic technique, i.e., the particle swarm algorithm. The Euler–Lagrange equations and the Pontryagin minimum principle are employed to express the control as a function of the adjoint variables conjugate to the dynamics equations.

The method that is being presented requires a reduced deal of effort in programming, debugging, and testing the algorithmic codes, as existing routines are used, in conjunction with analytical developments and a simple implementation of swarming algorithm. Hence, the methodology treated in this paper is intended to: (i) yield a reasonable solution for performance evaluation of multistage rockets and (ii) represent a technique for generating a suitable first-attempt guess trajectory to be employed by more refined algorithms tailored to optimizing the overall trajectory.

2. Problem definition

This research addresses the problem of performance evaluation of multistage rockets through the determination of a near optimal ascending trajectory, which terminates at injection of the payload in the desired operational orbit. This is a circular orbit of radius R_d, inclination i_d, and right ascension of the ascending node (RAAN) Ω_d. The three-stage launch vehicle is modeled as a point mass, in the context of a three-degree-of-freedom problem.

2.1. Rocket characteristics

The three-stage rocket that is being considered is the MultiRole Air Launch Missile (Muralm) [21], which is a rocket specifically designed for airlaunch from fighter aircraft. It has specified structural, propulsive, and aerodynamic characteristics and is portrayed in Fig. 1.

For the sake of simplicity, the mass distribution of the launch vehicle can be described in terms of masses of subrockets: subrocket 1 is the entire rocket, including all the three stages, subrocket 2 is the launch vehicle after the first stage separation, subrocket 3 is the launch vehicle after separation of the first two stages, and therefore is represented by the third stage only. Let $m_0^{(i)}$ denote the initial mass of subrocket i. This mass $m_0^{(i)}$ is composed

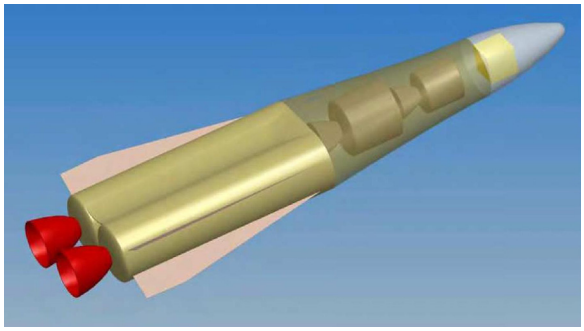

Fig. 1. Muralm rocket (conceptual design).

of a structural mass $m_S^{(i)}$, a propellant mass $m_P^{(i)}$, and a payload mass $m_U^{(i)}$ ($m_0^{(i)} = m_S^{(i)} + m_P^{(i)} + m_U^{(i)}$). For the first two subrockets $m_U^{(i)}$ ($i=1,2$) coincides with the initial mass of the subsequent subrocket (i.e., $m_U^{(i)} \equiv m_0^{(i+1)}$). With regard to the third subrocket $m_U^{(3)}$ is the (actual) payload mass, which is to be maximized. The mass distribution for the Muralm three-stage rocket is the following:

$$m_0^{(1)} = 3608 \text{ kg} \quad m_s^{(1)} = 306 \text{ kg} \quad m_p^{(1)} = 2480 \text{ kg} \quad m_U^{(1)} = 822 \text{ kg}$$
$$m_0^{(2)} = 822 \text{ kg} \quad m_s^{(2)} = 46 \text{ kg} \quad m_p^{(2)} = 370 \text{ kg} \quad m_U^{(2)} = 406 \text{ kg} \tag{1}$$

The initial mass of the third stage is specified and this implies that minimizing the third stage propellant expenditure implies maximizing the payload mass that can be injected in the desired orbit.

The propulsive characteristics of the launch vehicle can be described in terms of thrust magnitude T and specific impulse I_{SP}. Both these quantities are assumed as time-independent for all of the stages. If the superscript "$^{(j)}$" ($j=1,2,3$) refers to stage j, the following propulsive data characterize the Muralm rocket:

$$T^{(1)} = 235180 \text{ N} \quad T^{(2)} = 23472 \text{ N} \quad T^{(3)} = 7500 \text{ N}$$
$$I_{SP}^{(1)} = 290 \text{ sec} \quad I_{SP}^{(2)} = 291 \text{ sec} \quad I_{SP}^{(3)} = 320 \text{ sec} \tag{2}$$

For each subrocket i it is then straightforward to derive the thrust acceleration as a function of the time t,

$$\left(\frac{T}{m}\right)_i = \frac{n_0^{(i)}}{1 - n_0^{(i)}(t - t_{ign}^{(i)})/(g_0 I_{SP}^{(i)})} \tag{3}$$

where $n_0^{(i)}$ is the thrust acceleration at ignition of stage i (which occurs at the time $t_{ign}^{(i)}$) and g_0 is the gravitational acceleration at sea level. Lastly, due to the specified mass distribution, the burnout times $\Delta t_{BO}^{(i)}$ for the first two subrockets (i=1,2) are given by

$$\Delta t_{BO}^{(i)} = \frac{g_0 I_{SP}^{(i)}}{n_0^{(i)}} \frac{m_P^{(i)}}{m_0^{(i)}} \tag{4}$$

The aerodynamics of the Muralm rocket has been modeled through the Missile DATCOM [22] software. The aerodynamic force is assumed to be composed of two terms, lift L and drag D,

$$L = \frac{1}{2} c_L(\alpha, M) S^{(i)} \rho_A v^2 \quad \text{and} \quad D = \frac{1}{2} c_D(\alpha, M) S^{(i)} \rho_A v^2 \tag{5}$$

where the lift and drag coefficients c_L and c_D depend on M and α, i.e., Mach number and aerodynamic angle of attack, respectively. $S^{(i)}$ is the reference surface area for subrocket i, ρ_A is the (local) atmospheric density, and v is the relative velocity (cf. Appendix A). For the Muralm launch vehicle $S^{(j)} = 0.636 \text{ m}^2$ ($j=1,2,3$).

2.2. Equations of motion

The equations of motion that govern the rocket dynamics can be conveniently written in terms of its radius r, (geographical) longitude ξ, latitude ϕ, flight-path angle γ, velocity v, and heading angle ζ. These variables refer to the relative motion in an Earth-centered rotating frame (ECEF) and are described in more detail in Appendix A. They form the state vector $\mathbf{x}_R (\triangleq [r \quad \xi \quad \phi \quad \gamma \quad v \quad \zeta]^T)$ of the launch vehicle (in rotating coordinates). Omitting the superscript "$^{(i)}$", for each subrocket the equations of motion are

$$\dot{r} = v \sin\gamma \tag{6}$$

$$\dot{\xi} = \frac{v \cos\gamma \cos\zeta}{r \cos\phi} \tag{7}$$

$$\dot{\phi} = \frac{v \cos\gamma \sin\zeta}{r} \tag{8}$$

$$\dot{\gamma} = \frac{T}{m}\frac{\sin\alpha_T}{v} + \left(\frac{v}{r} - \frac{\mu_E}{r^2 v}\right)\cos\gamma + \frac{L}{mv} + \cos\phi\left[2\omega_E \cos\zeta + \frac{\omega_E^2 r}{v}(\cos\phi\cos\gamma + \sin\phi\sin\gamma\sin\zeta)\right] \tag{9}$$

$$\dot{v} = \frac{T}{m}\frac{\cos\alpha_T}{v} - \frac{\mu_E}{r^2}\sin\gamma - \frac{D}{m} + \omega_E^2 r \cos\phi(\cos\phi\sin\gamma - \sin\phi\cos\gamma\sin\zeta) \tag{10}$$

$$\dot{\zeta} = -\frac{v}{r}\tan\phi\cos\gamma\cos\zeta + 2\omega_E\cos\phi\tan\gamma\sin\zeta - \frac{\omega_E^2 r}{v\cos\gamma}\sin\phi\cos\phi\cos\zeta - 2\omega_E\sin\phi \tag{11}$$

where $\mu_E (= 398604.3 \text{ km}^3/\text{sec}^2)$ and $\omega_E (= 7.292115 \times 10^{-5} \text{ s}^{-1})$ are the Earth gravitational parameter and rotation rate, respectively, L and D are given by Eq. (5), and the time-varying mass is $m = m_0^{(i)} - T^{(i)}(t - t_{ign}^{(i)})/(g_0 I_{SP}^{(i)})$. As the thrust vector is assumed to be coplanar with the position vector $\boldsymbol{r}$ and the velocity vector $\boldsymbol{v}$, the angle α_T suffices to define its direction, which is taken clockwise from $\boldsymbol{v}$. The state $\boldsymbol{x}_R$ is continuous across stage separations, which occur at time t_1 (first stage separation) and t_2 (second stage separation).

Since the Muralm is assumed to be air-launched from a Tornado fighter, the initial conditions for Eqs. (6)–(11) are the following:

$$r(0) = R_E + h_0 \quad \xi(0) = \mu_L \quad \phi(0) = \phi_L \quad \gamma = \gamma_0 \quad v(0) = v_0$$
$$\zeta(0) = \frac{\pi}{2} - \chi_L \tag{12}$$

where h_0, γ_0, and v_0 are the rocket altitude, flight-path angle and (relative) velocity at release from the carrier aircraft. ϕ_L, χ_L, μ_L represent, respectively, the latitude, azimuth, and (geographical) longitude at release. $R_E (= 6378.165 \text{ km})$ is the Earth radius. For the problem

at hand the following data are assumed:

$$h_0 = 6.7 \text{ km} \quad \mu_L = -52.6 \text{ deg}$$
$$\phi_L = 5.2 \text{ deg} \quad \gamma_0 = 35.0 \text{ deg} \quad v_0 = 0.280 \text{ km/s} \tag{13}$$

The azimuth at release χ_L is determined in relation with the desired orbit inclination (cf. next section).

3. Method of solution

This work employs two distinct approaches for determining the trajectory of the first two stages and that of the last stage. The orbital plane selection is assumed to occur during the powered motion of the first two sub-rockets. Moreover, for the upper stage aerodynamic forces are negligible, and therefore they are not considered. These circumstances imply that the trajectory of the last stage is planar.

3.1. Trajectory of the first two stages

The determination of the trajectory of the first two stages is based on maximizing the final velocity at each stage separation, at which the flight-path angle is constrained to a specified value. The control is represented by the thrust direction, defined by α_T. However, in the context of trajectory analysis and optimization the thrust direction can be assumed as always aligned with the rocket longitudinal axis, and therefore these two angles coincide: $\alpha_T \equiv \alpha$.

While $\gamma(t_1)$ is set to 30 deg, the flight-path angle at the second stage separation, $\gamma(t_2)$, is adjusted iteratively up to obtaining an apogee altitude in the range $[h_d - \Delta h, h_d]$, where h_d is the altitude of the desired circular orbit, and Δh is set to 10 km. The first-attempt value for $\gamma(t_2)$ is chosen to be 25 deg.

The orbit inclination is selected through a simultaneous iterative process on the initial azimuth at rocket release. This process is relatively straightforward. In fact, in the range $[0, \pi/2]$ the orbit inclination increases as χ_L decreases (for $\chi_L \le \pi/2$). Convergence is declared when inclination is in the range $[i_d - \Delta i, i_d + \Delta i]$, where Δi is set to 3 deg. At each iteration of the process (for given values of $\gamma(t_1)$, $\gamma(t_2)$, and χ_L), the MATLAB routine *fmincon* determines the angle $\alpha(t)$ that maximizes v at each stage separation.

Finally, the correct selection of the release time t_L leads to obtaining the desired right ascension of the ascending node (RAAN), Ω_d. In fact, t_L is directly related to the Greenwich sidereal time at release θ_{gL},

$$\theta_{gL} = \overline{\theta}_g + \omega_E (t_L - \overline{t}) \tag{14}$$

where $\overline{\theta}_g$ is the (known) Greenwich absolute longitude at a reference time $\overline{t}$. The RAAN is given by the relations reported in Appendix B and, once the rocket final state (at orbit injection) is known, specifying the RAAN leads to obtaining θ_{gL} and then t_L,

$$t_L = \overline{t} + \frac{\theta_{gL} - \overline{\theta}_g + 2k\pi}{\omega_E} \qquad (k \in \mathbf{Z}) \tag{15}$$

At separation of the second stage, the state is denoted with $\boldsymbol{x}_R(t_2)$. Through the relations reported in Appendix B $\boldsymbol{x}_R(t_2)$ is employed to derive the inertial state, denoted with $\boldsymbol{x}_I(t_2)$ and including the following components: radius r, absolute longitude ξ_I, latitude ϕ, inertial flight-path angle γ_I, inertial velocity v_I, and inertial heading angle ζ_I. Then $\boldsymbol{x}_I(t_2)$ is employed to derive the orbital elements at t_2, i.e., the orbit semi-major axis (SMA) a_2, eccentricity e_2, inclination i, RAAN Ω, argument of perigee ω_2, and true anomaly $f(t_2)$ (cf. Appendix B).

3.2. Trajectory of the last stage

As remarked previously, the last stage trajectory is assumed to be planar. This circumstance implies that the orbit inclination and RAAN do not change during the last stage propulsion.

The third stage trajectory is assumed to be composed of two phases: a coast arc and a thrust phase. It is relatively straightforward to prove that during the coast arc the true anomaly variation, denoted with Δf, suffices to describe the rocket dynamics. In fact, if t_3 represents the ignition time of the third state, then $f_3 \triangleq f(t_3) = f(t_2) + \Delta f$. The orbital elements a_2, e_2, and ω_2 do not vary along a coast arc. As a result the radius, inertial velocity and inertial flight-path angle at t_3 are simply given by

$$r(t_3) = \frac{a_2(1-e_2^2)}{1+e_2 \cos f_3} \tag{16}$$

$$\gamma_I(t_3) = \arctan\left[\frac{e_2 \sin f_3}{1+e_2 \cos f_3}\right] \tag{17}$$

$$v_I(t_3) = \sqrt{\frac{\mu_E}{a_2(1-e_2^2)}}\sqrt{1+e_2^2+2e_2 \cos f_3} \tag{18}$$

In the powered phase, the third stage motion can be conveniently described through the use of the equations of motion for r, v_I, and γ_I (also referred to as the state equations hence forward),

$$\dot{r} = v_I \sin\gamma_I \quad \dot{v}_I = \frac{T}{m}\frac{\cos\alpha_I}{v_I} - \frac{\mu_E}{r^2}\sin\gamma_I \quad \dot{\gamma}_I = \frac{T}{m}\frac{\sin\alpha_I}{v_I} + \left(\frac{v_I}{r} - \frac{\mu_E}{r^2 v_I}\right)\cos\gamma_I \tag{19}$$

The thrust direction is now specified through the thrust angle α_I, taken clockwise from the direction of the inertial velocity vector $\boldsymbol{v}_I$. The operational orbit is assumed to a be circular orbit of radius R_d. This means that the final conditions at orbit injection (which occurs at t_f) are

$$r(t_f) = R_d \quad v_I(t_f) = \sqrt{\frac{\mu_E}{R_d}} \quad \gamma_I(t_f) = 0 \tag{20}$$

The trajectory for the last stage is obtained by minimizing fuel consumption, which is equivalent to minimizing the thrust duration. Hence, the objective function for the third stage trajectory optimization is

$$J = t_f - t_3 \tag{21}$$

In summary, the optimization problem for the upper stage is the following: determine the optimal control law $\alpha_I(t)$ and the optimal true anomaly f_3 such that J is minimized. The ignition time t_3 is then calculated through

the Kepler's law,

$$t_3 = t_2 + \sqrt{\frac{a_2^3}{\mu_E}}\{E(t_3) - E(t_2) - e_2 \sin[E(t_3)] + e_2 \sin[E(t_2)]\} \quad (22)$$

where $E(t_k)$ is the eccentric anomaly associated with $f(t_k)(k=2,3)$.

Letting $\boldsymbol{x} \triangleq [x_1 \quad x_2 \quad x_3]^T = [r \quad v_l \quad \gamma_l]^T$, to obtain the necessary conditions for an optimal solution a Hamiltonian H and a function of the boundary conditions Φ are introduced as

$$H \triangleq \lambda_1 x_2 \sin x_3 + \lambda_2 \left[\frac{T}{m}\frac{\cos\alpha_l}{x_2} - \frac{\mu_E}{x_1^2}\sin x_3\right] + \lambda_3 \left[\frac{T}{m}\frac{\sin\alpha_l}{x_2} + \left(\frac{x_2}{x_1} - \frac{\mu_E}{x_1^2 x_2}\right)\cos x_3\right] \quad (23)$$

$$\Phi \triangleq (t_f - t_3) + \upsilon_1\left[x_{10} - \frac{a_2(1-e_2^2)}{1+e_2\cos f_3}\right] + \upsilon_2\left[x_{20} - \sqrt{\frac{\mu_E}{a_2(1-e_2^2)}}\sqrt{1+e_2^2+2e_2\cos f_3}\right]$$
$$+\upsilon_3\left[x_{30} - \arctan\frac{e_2 \sin f_3}{1+e_2\cos f_3}\right] + \upsilon_4[x_{1f} - R_d]$$
$$+\upsilon_5\left[x_{2f} - \sqrt{\frac{\mu_E}{R_d}}\right] + \upsilon_6 x_{3f} \quad (24)$$

where $x_{k0} = x_k(t_3)$ and $x_{kf} = x_k(t_f)(k=1,2,3)$; $\boldsymbol{\lambda}(\triangleq [\lambda_1 \quad \lambda_2 \quad \lambda_3]^T)$ and $\boldsymbol{\upsilon}(\triangleq [\upsilon_1 \quad \upsilon_2 \quad \upsilon_3 \quad \upsilon_4 \quad \upsilon_5 \quad \upsilon_6]^T)$ represent, respectively, the adjoint variable conjugate to the dynamics Eq. (19) and to the boundary conditions (16)–(18) and (20). The necessary conditions for optimality [23] yield the following adjoint equations for the costate $\boldsymbol{\lambda}(\triangleq [\lambda_1 \quad \lambda_2 \quad \lambda_3]^T)$:

$$\dot{\lambda}_1 = (x_2\lambda_3\cos x_3)\frac{1}{x_1^2} - \left(2\mu_E\lambda_2\sin x_3 + 2\mu_E\lambda_3\frac{\cos x_3}{x_2}\right)\frac{1}{x_1^3} \quad (25)$$

$$\dot{\lambda}_2 = -\lambda_1\sin x_3 - \lambda_3\left[\cos x_3\left(\frac{1}{x_1} + \frac{\mu_E}{x_1^2 x_2^2}\right) - \frac{T_P}{m_P x_2^2}\sin\alpha_l\right] \quad (26)$$

$$\dot{\lambda}_3 = -x_2\lambda_1\cos x_3 + \mu_E\lambda_2\frac{\cos x_3}{x_1^2} + \lambda_3\sin x_3\left(\frac{x_2}{x_1} - \frac{\mu_E}{x_1^2 x_2}\right) \quad (27)$$

in conjunction with the respective boundary conditions,

$$\lambda_{k0} = -\upsilon_k \quad \text{and} \quad \lambda_{kf} = \upsilon_{k+3} \quad (k=1,2,3) \quad (28)$$

In the presence of initial conditions depending on a parameter (f_3) a pair of additional necessary conditions must hold [24],

$$\frac{\partial\Phi}{\partial f_3} = 0 \quad \text{and} \quad \frac{\partial^2\Phi}{\partial f_3^2} \geq 0 \quad (29)$$

The first equation yields a relation that expresses λ_{30} as a function of λ_{10}, λ_{20}, and f_3,

$$\lambda_{30} = \lambda_{20}\frac{\sin f_3\sqrt{1+e_2^2+2e_2\cos f_3}}{e_2+\cos f_3} - \lambda_{10}\frac{a_2(1-e_2^2)}{e_2+\cos f_3}\frac{1+e_2^2+2e_2\cos f_3}{(1+e_2\cos f_3)^2} \quad (30)$$

In addition, the optimal control α_l^* can be expressed as a function of the costates through the Pontryagin minimum principle

$$\alpha_l^* = \underset{\alpha_l}{\operatorname{argmin}} H \Rightarrow \sin\alpha_l^* = -\frac{\lambda_3}{x_2}\left[\left(\frac{\lambda_3}{x_2}\right)^2 + \lambda_2^2\right]^{-1/2} \text{ and}$$

$$\cos\alpha_l^* = -\lambda_2\left[\left(\frac{\lambda_3}{x_2}\right)^2 + \lambda_2^2\right]^{-1/2} \quad (31)$$

Lastly, as the final time is unspecified, the following transversality condition must hold:

$$H(t_f) + \frac{\partial\Phi}{\partial t_f} = 0 \Rightarrow \frac{n_0^{(3)}}{1 - n_0^{(3)}(t_f - t_3)/(g_0 I_{SP}^{(3)})}\sqrt{\left[\frac{\lambda_{3f}}{x_{2f}}\right]^2 + [\lambda_{2f}]^2} - 1 = 0 \quad (32)$$

The necessary conditions for optimality allow translating the optimal control problem into a two-point boundary-value problem involving Eqs. (25)–(32), with unknowns represented by the initial values of $\boldsymbol{\lambda}$, f_3, and t_f.

Equality constraints reduce the search space where feasible solutions can be located. However, what will be demonstrated is that for the problem at hand the transversality condition can be neglected by the swarming algorithm. To do this, one has first to recognize the special structure of the costate Eqs. (25)–(27), which are homogeneous in λ. This circumstance implies that if an optimization algorithm is capable of finding some initial value of λ such that $\lambda_1(0) = k_\lambda\lambda_1^*(0), \lambda_2(0) = k_\lambda\lambda_2^*(0)$, and $\lambda_3(0) = k_\lambda\lambda_3^*(0)$ $(k_\lambda > 0)$ (where the superscript "*" denotes the actual optimal value of a variable), then the same proportionality holds between $\boldsymbol{\lambda}$ and $\boldsymbol{\lambda}^*$ at any t, due to homogeneity of Eqs. (25)–(27). Moreover, the control α_l can be written as a function of $\boldsymbol{\lambda}$ through Eq. (31) and one can recognize that α_l coincides with α_l^*,

$$\sin\alpha_l = -\frac{\lambda_3}{x_2}\left[\left(\frac{\lambda_3}{x_2}\right)^2 + \lambda_2^2\right]^{-1/2} = -\frac{k_\lambda\lambda_3^*}{x_2}\left[\left(\frac{k_\lambda\lambda_3^*}{x_2}\right)^2 + (k_\lambda\lambda_2^*)^2\right]^{-1/2} \equiv \sin\alpha_l^* \quad (33)$$

$$\cos\alpha_l = -\lambda_2\left[\left(\frac{\lambda_3}{x_2}\right)^2 + \lambda_2^2\right]^{-1/2} = -k_\lambda\lambda_2^*\left[\left(\frac{k_\lambda\lambda_3^*}{x_2}\right)^2 + (k_\lambda\lambda_2^*)^2\right]^{-1/2} \equiv \cos\alpha_l^* \quad (34)$$

This circumstance implies that if $\boldsymbol{\lambda}$ is proportional to $\boldsymbol{\lambda}^*$ then the final conditions are fulfilled at the minimum final time t_f^*. In contrast, the transversality condition is violated, because the value of $H(t_f^*)$, due to Eq. (32), turns out to be

$$H(t_f^*) = -\frac{n_0^{(3)}}{1 - n_0^{(3)}(t_f^* - t_3^*)(g_0 I_{SP}^{(3)})}\sqrt{\left[\frac{\lambda_{3f}}{x_{2f}}\right]^2 + [\lambda_{2f}]^2} = -k_\lambda \neq -1 \quad (35)$$

Therefore, provided that the proportionality condition holds, the optimal control α_l^* can be determined without considering the transversality condition, which in fact is ignorable in this context. Thus, this condition is discarded, in order to reduce the number of equality constraints considered by the swarming algorithm, with the intent of improving its performance.

In short, the following parameter set can be employed in the solution process: $\{\lambda_{10}, \lambda_{20}, f_3, t_f\}$. The remaining parameter λ_{30} can be easily obtained by means of the relation (30). Specifically, the swarming technique is based on the following points:

(a) the control variable is expressed as a function of the costate through Eq. (31)

(b) the value of λ_{30} is calculated by means of Eq. (30), after picking the unknown values of the remaining Lagrange multipliers at the initial time (λ_{10} and λ_{20}), and the true anomaly f_3
(c) the state Eq. (19) are integrated numerically up to the final time t_f, together with the adjoint Eqs. (25)–(27). The respective initial conditions are known once the parameters f_3 (for the state equations) and $\{\lambda_{10},\lambda_{20},\lambda_{30}\}$ (for the adjoint equations) are specified
(d) the inequality in Eq. (29) (not expanded for the sake of brevity) and the conditions at injection (20) are evaluated.

In summary, the problem reduces to the determination of four unknown parameters, $\{\lambda_{10},\lambda_{20},f_3,t_f\}$, that lead the dynamical system to satisfying three conditions (Eq. (20)).

3.3. Outline of the particle swarm optimization method

The particle swarm optimization (PSO) method is a heuristic technique aimed at finding the optimal values of a set of unknown parameters, for a generic dynamical system. In this research the optimization problem involves a continuous time-dependent control variable and is translated into a parameter optimization problem through the necessary conditions for optimality, which allow expressing the control variable as a function of the adjoint variables conjugate to the dynamics equations.

Unconstrained parameter optimization problems can be stated as follows: determine the optimal values of the n unknown parameters $\{\chi_1,\ldots,\chi_n\}$ such that the objective function J is minimized. The PSO technique is a population-based method, where the population is represented by a swarm of N particles. Each particle i ($i=1,\ldots,N$) is associated with a position vector $\boldsymbol{\chi}(i)$ and with a velocity vector $\boldsymbol{w}(i)$. The position vector includes the values of the n unknown parameters of the problem, $\boldsymbol{\chi}(i)\triangleq\begin{bmatrix}\chi_1(i) & \cdots & \chi_n(i)\end{bmatrix}^T$, whereas the velocity vector, whose components are denoted with $w_k(i)$ ($k=1,\ldots,n$), determines the position update. Both the position and the velocity components are constrained to suitable ranges,

$$a_k \le \chi_k \le b_k \quad \text{and} \quad -d_k \le w_k \le d_k \quad (k=1,\ldots,n) \tag{36}$$

where $d_k \triangleq b_k - a_k$. Each particle represents a possible solution to the problem, and corresponds to a specific value of the objective function. The initial population is randomly generated by introducing N particles, whose positions and velocities are (stochastically) uniformly distributed in the respective search spaces, defined by Eq. (36). The expressions for position and velocity update determine the swarm evolution toward the location of the globally optimal position, which corresponds to the globally optimal solution to the problem of interest. The algorithm terminates when the maximum number of iterations N_{IT} is reached or an alternative convergence criterion is met. The position vector of the best particle is expected to contain the optimal values of the unknown parameters, which correspond to the global minimum of J. A fairly large number of iterations are set to ensure that the final solution is stable enough to be considered optimal. The central idea underlying the method is contained in the formula for velocity updating. This formula includes three terms with stochastic weights: the first term is the so called inertial component and for each particle is proportional to its velocity in the preceding iteration; the second component is termed the cognitive component, directed toward the personal best position, i.e., the best position experienced by the particle; and finally the third term is the social component, directed toward the global best position, i.e., the best position yet located by any particle in the swarm. A detailed description of the algorithm (with also the same settings adopted in this work) is contained in [25].

Constrained optimization problems involve equalities and/or inequalities, regarding (directly or indirectly) the unknown parameters. Equality constraints narrow considerably the search space where feasible solutions can be located. This is due to the fact that (non-redundant) equality constraints actually reduce the degree of freedom of the problem according to their number. In fact, m equality constraints reduce the degree of freedom by m. Therefore, in the presence of n unknown parameters, at most $m=n$ equality constraints are admissible ($m\le n$): $l_r(\boldsymbol{\chi})=0$ ($r=1,\ldots,m$). The most popular approach for dealing with these constraints consists in penalizing them by adjoining additional terms to the objective function,

$$\tilde{J}=J+\sum_{r=1}^{m}\eta_r\,|l_r(\boldsymbol{\chi})| \tag{37}$$

This approach is employed also in this research, with $\eta_r=1$ ($r=1,2,3$). The equality constraints are represented by the relations (20). As the necessary conditions for optimality are used to express the control variable, the PSO assumes as objective the expression (37) with $J\equiv 0$. Inequality constraints are less problematic because they reduce the search space of feasible solutions without decreasing the degree of freedom of the problem. For each particle the simplest way of treating inequality constraints consists in assigning a fictitious infinite value to the objective function if the particle violates at least one of them. In this work a single inequality constraint exists, i.e., the second relation reported in Eq. (29). Further options exist for dealing with constraints (e.g., methods based on preserving feasibility of solutions, methods that distinguish between feasible and infeasible solutions, cf. [26,27]). However, the simple strategy described previously for equality and inequality constraints will be shown to be quite effective for the problem considered in this paper.

4. Ascending trajectories

The ascending trajectories are determined by employing canonical units: the distance unit (DU) is the Earth radius, whereas the time unit (TU) is such that $\mu_E=1$ DU^3/TU^2. Thus, 1 DU=6378.165 km and 1 TU=806.8 sec.

With regard to the last stage trajectory, the swarming optimizer is employed with the following settings: $N=50$ (number of particles) and $N_{IT}=500$ (maximum number of

iterations). The algorithm terminates prematurely its execution if the objective $\tilde{J}$ (cf. Eq. (37)) decreases below the value 10^{-6}. The optimal values of the unknown parameters are sought in the following ranges: $-1 \le \lambda_{k0} \le 1$ $(k=1,2,3)$, $0 \le \Delta f \le \pi$, and $0.01\ \text{TU} \le t_f \le 0.2\ \text{TU}$. It is worth noticing that the constraint reduction allows arbitrarily defining the search space for the initial values of the Lagrange multipliers. This means that they can be sought in the interval $-1 \le \lambda_{k0} \le 1$ by the PSO algorithm, and only a posteriori their correct values (fulfilling also the transversality condition (32)) can be recovered.

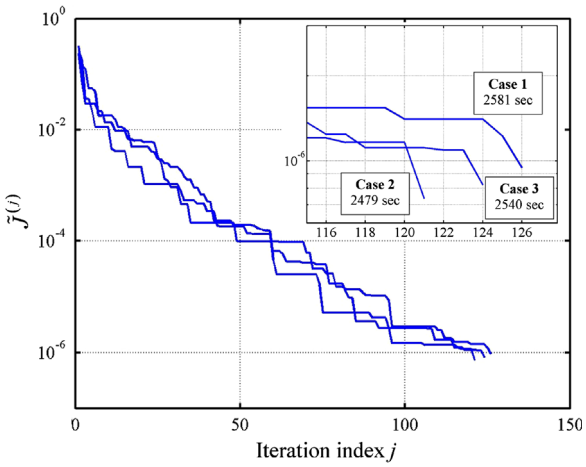

Fig. 2. Objective evolution as a function of the iteration index (computational times are indicated in the insets).

Three test cases have been considered:

Case 1 : $h_d = 280$ km and $i_d = 5.2$ deg
Case 2 : $h_d = 280$ km and $i_d = 45$ deg
Case 3 : $h_d = 280$ km and $i_d = 90$ deg

Fig. 2 shows the objective evolution as a function of the iteration index for all of the three cases, together with the runtimes (on an AMD Sempron Processor 3200). The main results (azimuth at release χ_L, flight-path angle at the second stage separation $\gamma(t_2)$, orbit semi-major axis a_f, eccentricity e_f, inclination i_f, coast duration Δt_{CO}, and final mass at injection m_f) are reported in the following.

$$\textbf{\textit{Case}}\ 1: \quad \chi_L = 90.0\ \text{deg} \quad \gamma(t_2) = 20.0\ \text{deg} \quad \Delta t_{CO} = 175.2\ \text{s}$$
$$a_f = R_E + 280.004\ \text{km} \quad e_f = 7.260 \cdot 10^{-7} \quad i_f = 5.2\ \text{deg} \quad m_f = 140.4\ \text{kg} \tag{38}$$

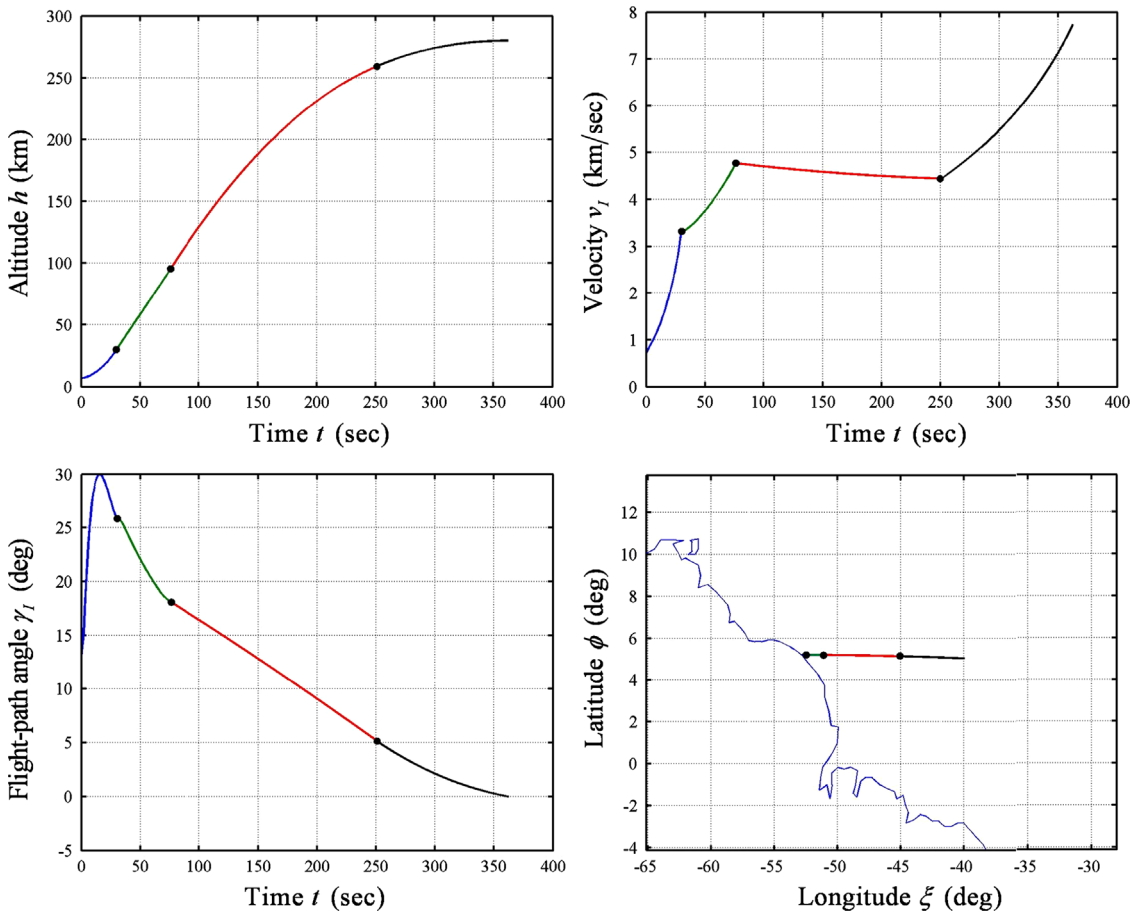

Fig. 3. Case 1: altitude, inertial velocity and flight-path angle, ground track.

$$\textbf{Case } 2: \quad \chi_L = 40.0 \text{ deg} \quad \gamma(t_2) = 20.5 \text{ deg} \quad \Delta t_{CO} = 171.3 \text{ s}$$
$$a_f = R_E + 280.002 \text{ km} \quad e_f = 4.377 \cdot 10^{-7} \quad i_f = 45.7 \text{ deg} \quad m_f = 133.7 \text{ kg} \tag{39}$$

$$\textbf{Case } 3: \quad \chi_L = -4.0 \text{ deg} \quad \gamma(t_2) = 21.0 \text{ deg} \quad \Delta t_{CO} = 158.2 \text{ s}$$
$$a_f = R_E + 279.997 \text{ km} \quad e_f = 4.403 \cdot 10^{-7} \quad i_f = 87.7 \text{ deg} \quad m_f = 120.6 \text{ kg} \tag{40}$$

From inspecting the results summarized in Eqs. (38)–(40) it is apparent that the final conditions at injection are fulfilled in a satisfactory way. Moreover, the final mass decreases as the orbit inclination increases. This is a predictable result and is related to the launch site, which is located in the proximity of the Earth equator. This circumstance implies that as the operational

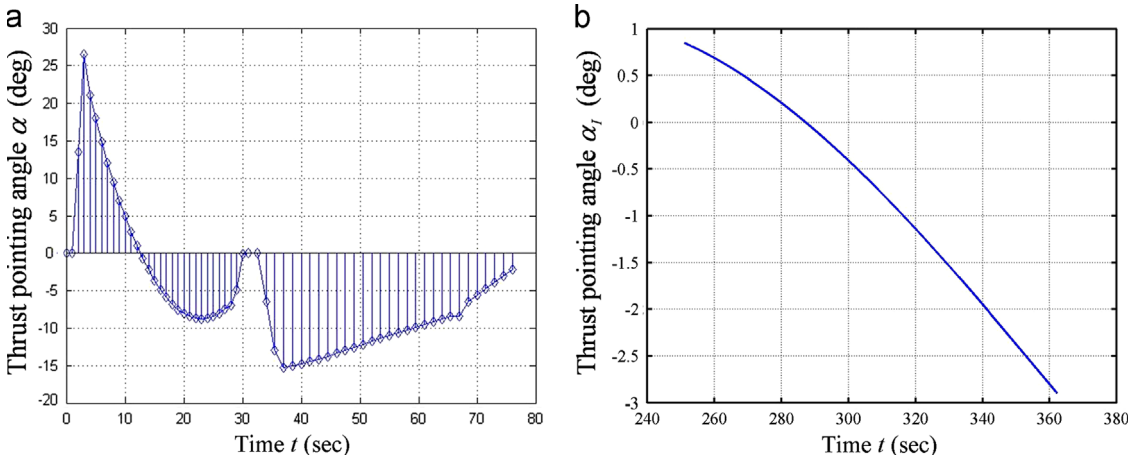

Fig. 4. Case 1: control for the first two stages (a) and for the last stage (b).

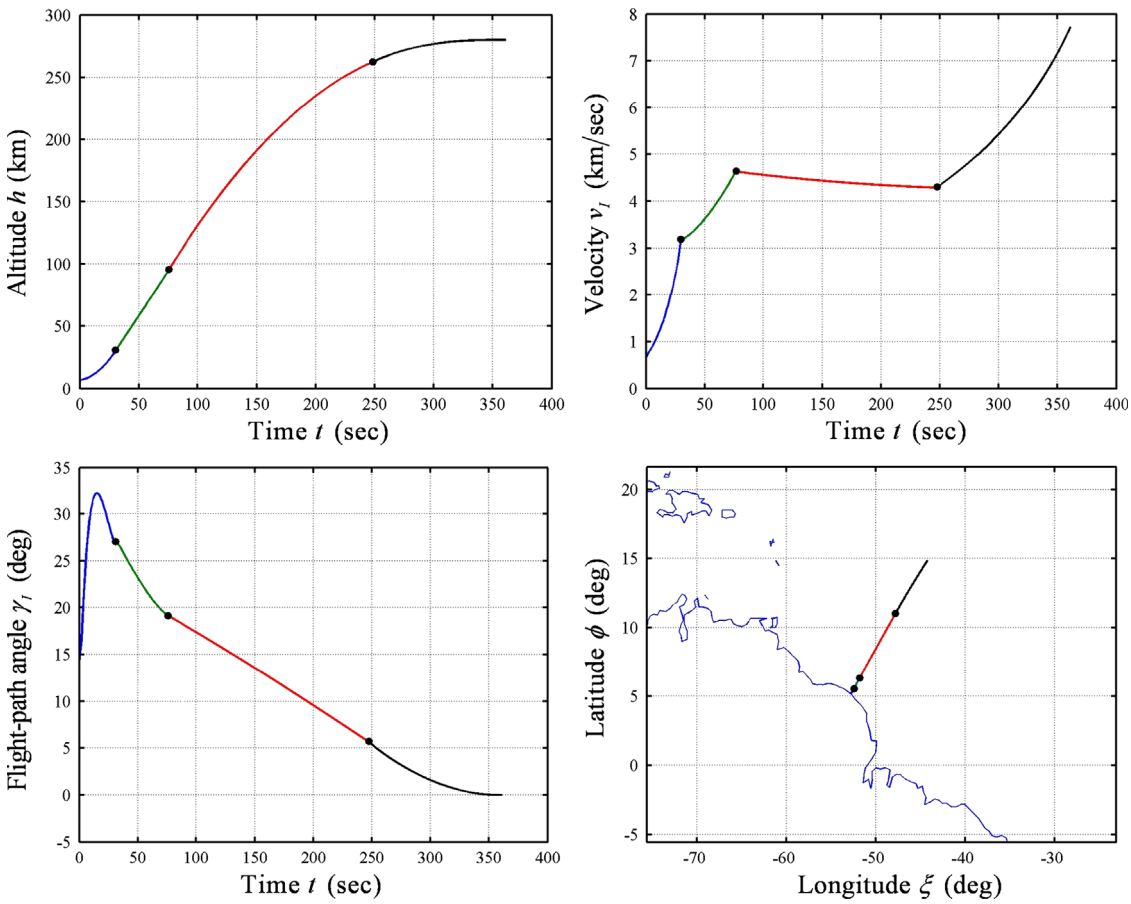

Fig. 5. Case 2: altitude, inertial velocity and flight-path angle, ground track.

orbit inclination increases, an increasing amount of fuel is employed to correct the initial inertial velocity, which has a nonnegligible component directed toward the East. Figs. 3, 5, and 8 portray some state components (radius, velocity and flight-path angle) and the ground tracks, whereas Figs. 4, 6, and 7 illustrate the corresponding thrust pointing angle time histories for each of the three subrockets.

Despite its simplicity with respect to alternative existing approaches, the technique applied in this work yields trajectories that resemble qualitatively the ones reported in [2,12,13,14,16,17,20] and those of existing rockets such as the Scout [28,29]. In fact, either for three-stage and for four-stage rockets, the upper stage trajectory is composed of two arcs: (i) a coast arc, in which the launch vehicle converts kinetic energy into potential energy by increasing its altitude, and (ii) a powered phase, in which a near-horizontal burn finally circularizes the orbit to the desired altitude.

5. Concluding remarks

The generation of an optimal trajectory for multistage launch vehicles is a challenging problem, treated with different approaches in the past. This work proposes and successfully applies a simple technique for generating near-optimal three-dimensional ascending trajectories for multistage rockets, for the purpose of performance evaluation. Only existing routines and a simple implementation of swarming algorithm are employed, in conjunction with the analytical necessary conditions for optimality, applied to the upper stage trajectory. A particular configuration of the launch vehicle is assumed, taking into account realistic propulsive, structural, and aerodynamic data.

This study proves how the desired orbit right ascension of the ascending node and inclination can be obtained by properly selecting the launch time and the azimuth angle at release, respectively. The remaining orbit elements of the operational orbit are attained by choosing a suitable control law for the thrust direction time history, and by selecting the optimal coast duration at the second stage separation.

The solutions found in this work resemble the optimal trajectories found in the past for alternative launch vehicles through different techniques, and appear suitable also for being employed as guesses for more refined optimization algorithms.

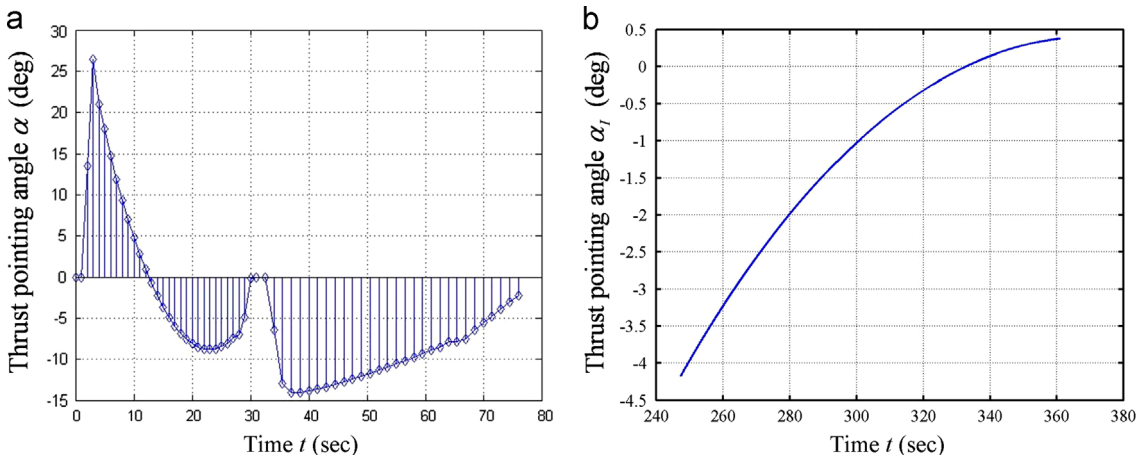

Fig. 6. Case 2: control for the first two stages (a) and for the last stage (b).

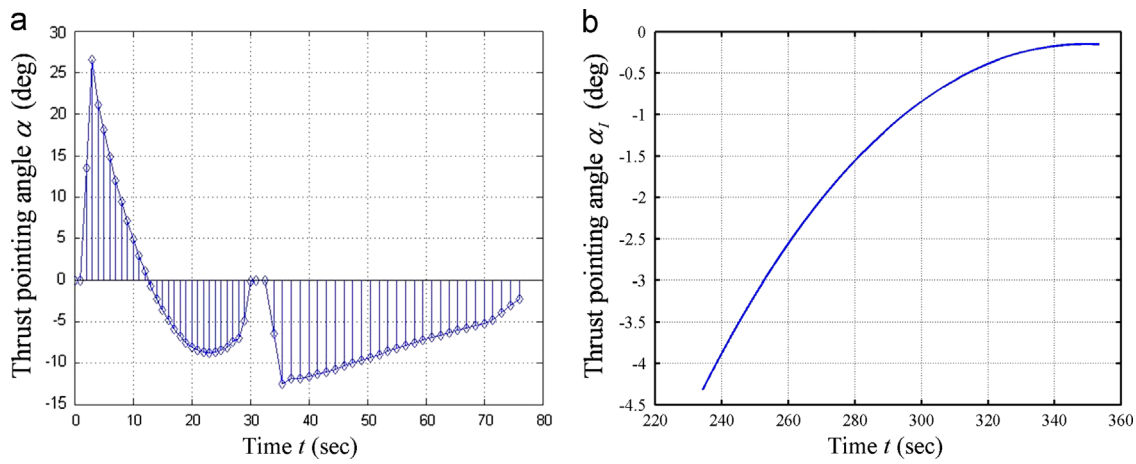

Fig. 7. Case 3: control for the first two stages (a) and for the last stage (b).

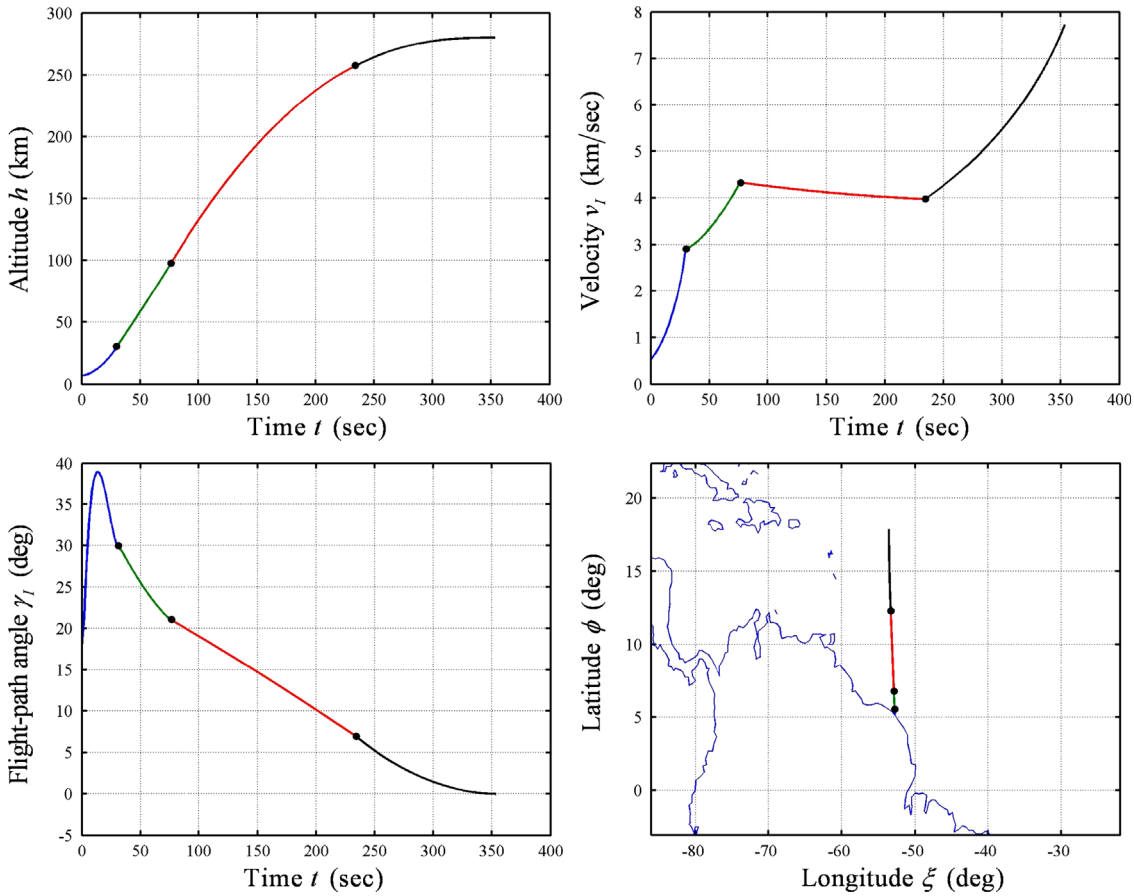

Fig. 8. Case 3: altitude, inertial velocity and flight-path angle, ground track.

Appendix A. Reference frames

The rocket motion is conveniently described in rotating, i.e., non-inertial, reference frames. The Earth-centered Earth-fixed (ECEF) reference frame represents a reference system that is rigidly attached to the Earth. Under the assumption that the planet rotates with the constant angular rate ω_E this circumstance implies that also the reference frame rotates with angular rate ω_E with respect to an inertial Earth-centered (ECI) reference frame, denoted with $(\hat{c}_1,\hat{c}_2,\hat{c}_3)$. The origin of both the inertial and the rotating coordinate system is the center of the planet, O; $\hat{c}_1$ is the vernal axis. The unit vector $\hat{c}_3 \equiv \hat{k}$ is aligned with the planet axis of rotation and is positive northward, and therefore $\omega_E\hat{k}$ ($\triangleq \boldsymbol{\omega}_E$) represents the (vector) rotation rate of the ECEF-frame with respect to the ECI-frame. The unit vector $\hat{\imath}$ intersects the Greenwich meridian at all times. The ECEF-frame is associated with $(\hat{\imath},\hat{\jmath},\hat{k})$, which form a right-handed, time-dependent sequence of unit vectors. As the reference Greenwich meridian rotates with rotation rate ω_E, its angular position (with respect to the ECI-frame) is identified by its absolute longitude (usually termed Greenwich sidereal time) $\theta_g(t)=\theta_g(\bar{t})+\omega_E(t-\bar{t})$, where $\bar{t}$ denotes a generic time instant.

The position vector of the orbiting spacecraft in the ECEF-frame is denoted with $\boldsymbol{r}$, whereas the subscript "I" corresponds to a quantity in the ECI-frame. As the ECEF-frame and the ECI-frame have a common origin, it is evident that $\boldsymbol{r}\equiv\boldsymbol{r}_I$. In contrast, the inertial velocity $\boldsymbol{v}_I$ is related to the (relative) velocity $\boldsymbol{v}$ through the following expression:

$$\boldsymbol{v}_I = \boldsymbol{v} + \boldsymbol{\omega}_E \times \boldsymbol{r} \tag{41}$$

This means that, unlike the position vector, the velocity vector in rotating coordinates, $\boldsymbol{v}$, does not coincide with the inertial velocity vector, $\boldsymbol{v}_I$.

An additional useful reference frame is represented by the right-handed Cartesian coordinate system $(\hat{r},\hat{E},\hat{N})$, where $\hat{r}$ is the unit vector aligned with $\boldsymbol{r}$, $\hat{E}$ is directed along the local East direction, and $\hat{N}$ is aligned with the local North direction. It is straightforward to obtain $(\hat{r},\hat{E},\hat{N})$ from $(\hat{\imath},\hat{\jmath},\hat{k})$ through two elementary rotations: the first (counterclockwise by the geographical longitude angle ξ) about axis 3, the second (clockwise by the latitude angle ϕ) about axis 2.

A third useful rotating frame is $(\hat{r},\hat{\theta},\hat{h})$, where $\hat{\theta}$ is aligned with the projection of the velocity $\boldsymbol{v}$ onto the local horizontal plane $(\hat{E},\hat{N})$. This frame is obtained from $(\hat{r},\hat{E},\hat{N})$ through a counterclockwise rotation about axis 1 by the

heading angle ζ (cf. Fig. 8). As a result, $(\hat{r},\hat{\theta},\hat{h})$ can be written as a function of $(\hat{\imath},\hat{\jmath},\hat{k})$: $\begin{bmatrix}\hat{r} & \hat{\theta} & \hat{h}\end{bmatrix}^T = \boldsymbol{R}_A\begin{bmatrix}\hat{\imath} & \hat{\jmath} & \hat{k}\end{bmatrix}^T$, where

$$\boldsymbol{R}_A = \begin{bmatrix} \cos\phi\cos\xi & \cos\phi\sin\xi & \sin\phi \\ -\sin\zeta\sin\phi\cos\xi-\cos\zeta\sin\xi & -\sin\zeta\sin\phi\sin\xi+\cos\zeta\cos\xi & \sin\zeta\cos\phi \\ -\cos\zeta\sin\phi\cos\xi+\sin\zeta\sin\xi & -\cos\zeta\sin\phi\sin\xi-\sin\zeta\cos\xi & \cos\zeta\cos\phi \end{bmatrix} \tag{42}$$

Finally, the reference frame $(\hat{n},\hat{v},\hat{h})$ (with $\hat{v}$ aligned with $\boldsymbol{v}$) is obtained from $(\hat{r},\hat{\theta},\hat{h})$ through a clockwise rotation about axis 3 by the flight-path angle γ (Fig. A1). From inspecting Fig. A1 it is apparent that

$$\boldsymbol{v} = v\begin{bmatrix}\sin\gamma & \cos\gamma\cos\zeta & \cos\gamma\sin\zeta\end{bmatrix}\begin{bmatrix}\hat{r} & \hat{E} & \hat{N}\end{bmatrix}^T \tag{43}$$

The overall aerodynamic force $\boldsymbol{A}$ is conveniently written in the $(\hat{n},\hat{v},\hat{h})$-frame as the sum of two terms, i.e., lift L and drag D:

$$\mathbf{A} = L\hat{n} - D\hat{v} \tag{44}$$

Appendix B. State variables and orbit elements

The derivation of the orbital elements (a,e,i,Ω,ω,f) from the relative state variables $(r,\xi,\phi,\gamma,v,\zeta)$ follows two steps:

(1) derivation of the inertial state variables $(r_I,\xi_I,\phi_I,\gamma_I,v_I,\zeta_I)$ from the relative state variables $(r,\xi,\phi,\gamma,v,\zeta)$
(2) derivation of the orbital elements (a,e,i,Ω,ω,f) from the inertial state variables $(r_I,\xi_I,\ \phi_I,\gamma_I,v_I,\zeta_I)$.

With regard to step (1), the fact that $\boldsymbol{r}\equiv\boldsymbol{r}_I$ and the definition of the reference $(\hat{\imath},\hat{\jmath},\hat{k})$ implies that

$$r_I = r \quad \xi_I = \xi+\theta_g \quad \phi_I = \phi \tag{45}$$

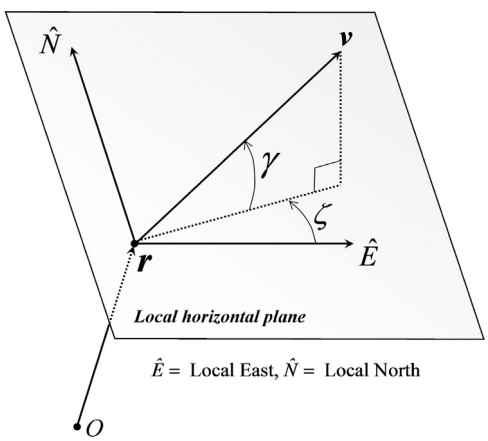

Fig. A1. $(\hat{r},\hat{E},\hat{N})$-frame, $(\hat{n},\hat{v},\hat{h})$-frame, and related angles.

The velocity vectors $\boldsymbol{v}_I$ and $\boldsymbol{v}$ have the following expressions in the rotating frame $(\hat{r},\hat{E},\hat{N})$:

$$\mathbf{v}_I = v_I\begin{bmatrix}\sin\gamma_I \\ \cos\gamma_I\cos\zeta_I \\ \cos\gamma_I\sin\zeta_I\end{bmatrix}^T\begin{bmatrix}\hat{r} \\ \hat{E} \\ \hat{N}\end{bmatrix} \quad \text{and} \quad \mathbf{v} = v\begin{bmatrix}\sin\gamma \\ \cos\gamma\cos\zeta \\ \cos\gamma\sin\zeta\end{bmatrix}^T\begin{bmatrix}\hat{r} \\ \hat{E} \\ \hat{N}\end{bmatrix} \tag{46}$$

Due to Eq. (46) and to the fact that $\boldsymbol{\omega}_E\times\boldsymbol{r} = \omega_E r\cos\phi\hat{E}$, Eq. (41) yields three relations:

$$v_I\sin\gamma_I = v\sin\gamma \tag{47}$$

$$v_I\cos\gamma_I\cos\zeta_I = v\cos\gamma\cos\zeta+\omega_E r\cos\phi \tag{48}$$

$$v_I\cos\gamma_I\sin\zeta_I = v\cos\gamma\sin\zeta \tag{49}$$

which lead to the following expressions of γ_I, v_I, and ζ_I as functions of γ, v, and ζ:

$$v_I = \sqrt{v^2+(\omega_E r\cos\phi)^2+2v\omega_E r\cos\phi\cos\gamma\cos\zeta} \tag{50}$$

$$\gamma_I = \arcsin\left[\frac{v\sin\gamma}{v_I}\right] \tag{51}$$

$$\cos\zeta_I = \frac{v\cos\gamma\cos\zeta+\omega_E r\cos\phi}{v_I\cos\gamma_I} \quad \text{and} \quad \sin\zeta_I = \frac{v\cos\gamma\sin\zeta}{v_I\cos\gamma_I} \tag{52}$$

With regard to step (2), the in-plane orbital elements a, e, and f can be easily calculated from r, γ_I, and v_I. In fact, the conservation of energy yields a:

$$a = \frac{\mu_E r}{2\mu_E - r v_I^2} \tag{53}$$

Then, the angular momentum magnitude equals $r v_I\cos\gamma_I$ and can also be written in terms of orbital elements as $\sqrt{\mu_E a(1-e^2)}$. Hence, the orbit eccentricity is given by

$$e = \sqrt{1-\frac{(r v_I\cos\gamma_I)^2}{\mu_E a}} \tag{54}$$

The true anomaly f can be found by considering the polar equation of the ellipse,

$$r = \frac{a(1-e^2)}{1+e\cos f} \rightarrow \cos f = \frac{a(1-e^2)-r}{re} \tag{55}$$

in conjunction with the radial component of velocity:

$$\sqrt{\frac{\mu}{a(1-e^2)}}e\sin f = v\sin\gamma \rightarrow \sin f = \frac{v\sin\gamma}{e}\sqrt{\frac{a(1-e^2)}{\mu}} \tag{56}$$

The remaining orbital elements (i, Ω, and ω) can be obtained by considering the rotation matrix that relates $(\hat{r},\hat{\theta},\hat{h})$ to $(\hat{c}_1,\hat{c}_2,\hat{c}_3)$, written in terms of the angles i, Ω, and $\theta\ (\triangleq\omega+f)$: $\begin{bmatrix}\hat{r} & \hat{\theta} & \hat{h}\end{bmatrix}^T = \boldsymbol{R}_B\begin{bmatrix}\hat{c}_1 & \hat{c}_2 & \hat{c}_3\end{bmatrix}^T$,

where [30]:

$$\boldsymbol{R}_B = \begin{bmatrix} \cos\theta\cos\Omega - \sin\theta\cos i\sin\Omega & \cos\theta\sin\Omega + \sin\theta\cos i\cos\Omega & \sin\theta\sin i \\ -\sin\theta\cos\Omega - \cos\theta\cos i\sin\Omega & -\sin\theta\sin\Omega + \cos\theta\cos i\cos\Omega & \cos\theta\sin i \\ \sin i\sin\Omega & -\sin i\cos\Omega & \cos i \end{bmatrix} \tag{57}$$

$(\hat{r},\hat{\theta},\hat{h})$ and $(\hat{c}_1,\hat{c}_2,\hat{c}_3)$ are also related through a rotation matrix $\tilde{\boldsymbol{R}}_A$ formally identical to $\boldsymbol{R}_A$ (cf. Eq. (42)), with ξ_I and ζ_I in place of ξ and ζ, respectively. The fact that $\tilde{\boldsymbol{R}}_A \equiv \boldsymbol{R}_B$ leads to the following relations:

$$i = \arccos(\cos\zeta_I\cos\phi) \tag{58}$$

$$\sin\Omega = \frac{-\cos\zeta_I\sin\phi\cos\xi_I + \sin\zeta_I\sin\xi_I}{\sin i}$$
$$\cos\Omega = \frac{\cos\zeta_I\sin\phi\sin\xi_I + \sin\zeta_I\cos\xi_I}{\sin i} \tag{59}$$

$$\sin\theta = \frac{\sin\phi}{\sin i} \quad \cos\theta = \frac{\sin\zeta_I\cos\phi}{\sin i} \tag{60}$$

and finally to obtaining $\omega = \theta - f$. Due to Eqs. (14), (15), and (59) the proper choice of the release time allows setting ξ_I to the value that corresponds to the desired RAAN Ω_d.

References

[1] A. Miele, Multiple-subarc gradient-restoration algorithm, Part 1: Algorithm structure, J. Optimiz. Theory Appl. 116 (1) (2003) 1–17.
[2] A. Miele, Multiple-subarc gradient-restoration algorithm, Part 1: Application to a multistage launch vehicle design, J. Optimiz. Theory Appl. 116 (1) (2003) 19–39.
[3] K.R. Brown, E.F. Harrold, G.W. Johnson, Rapid Optimization of Multiple-Burn Rocket Flights. NASA CR-1430, 1969.
[4] C.R. Hargraves, S.W. Paris, Direct trajectory optimization using nonlinear programming and collocation, J. Guid. Control Dynam. 10 (4) (1987) 338–342.
[5] A.L. Herman, B.A. Conway, Direct optimization using collocation based on high-order Gauss–Lobatto quadrature rules, J. Guid. Control Dynam. 19 (3) (1996) 592–599.
[6] P.J. Enright, B.A. Conway, Discrete approximations to optimal trajectories using direct transcription and nonlinear programming, J. Guid. Control Dynam. 15 (4) (1992) 994–1002.
[7] J.T. Betts, Optimal interplanetary orbit transfers by direct transcription, J. Astronaut. Sci. 42 (3) (1994) 247–326.
[8] H. Seywald, Trajectory optimization based on differential inclusion, J. Guid. Control Dynam. 17 (3) (1994) 480–487.
[9] V. Coverstone-Carroll, S.N. Williams, Optimal low thrust trajectories using differential inclusion concepts, J. Astronaut. Sci. 42 (4) (1994) 379–393.
[10] A.J. Calise, S. Tandon, D.H. Young, S. Kim Further improvements to a hybrid method for launch vehicle ascent trajectory optimization, in: AIAA Guidance, Navigation, and Control Conference and Exhibit, Denver, CO, 2000; Paper AIAA 2000-4261.
[11] P.F. Gath, A.J. Calise, Optimization of launch vehicle ascent trajectories with path constraints and coast arcs, J. Guid. Control Dynam. 24 (2) (2001) 296–304.
[12] P. Lu, B. Pan, Trajectory Optimization and Guidance for an Advanced Launch System. 30th Aerospace Sciences Meeting and Exhibit, Reno, NV, 1992; Paper AIAA 1992-732.
[13] P. Lu, B.J. Griffin, G.A. Dukeman, F.R. Chavez, Rapid optimal multi-burn ascent planning and guidance, J. Guid. Control Dynam. 31 (6) (2008) 1656–1664.
[14] N. Weigel, K.H. Well, Dual payload ascent trajectory optimization with a splash-down constraint, J. Guid. Control Dynam. 23 (1) (2000) 45–52.
[15] P. Martinon, F. Bonnans, J. Laurent-Varin, E. Trélat, Numerical study of optimal trajectories with singular arcs for an Ariane 5 launcher, J. Guid. Control Dynam. 32 (1) (2009) 51–55.
[16] W. Roh, Y. Kim, Trajectory optimization for a multi-stage launch vehicle using time finite element and direct collocation methods, Eng. Optim. 34 (1) (2002) 15–32.
[17] R. Jamilnia A. Naghash, Simultaneous optimization of staging and trajectory of launch vehicles using two different approaches. Aerosp. Sci. Technol., 10.1016/j.ast.2011.06.013.
[18] P.E. Gill, W. Murray, M.A. Saunders, User's Guide for SNOPT Version 7: Software for Large-Scale Nonlinear Programming. Stanford University, Stanford, CA.
[19] D.J. Bayley, R.J. Hartfield Jr., J.E. Burkhalter, R.M. Jenkins, Design optimization of a space launch vehicle using a genetic algorithm, J. Spacecraft Rockets 45 (4) (2008) 733–740.
[20] M.D. Qazi, H. Linshu, T. Elhabian. Rapid Trajectory Optimization Using Computational Intelligence for Guidance and Conceptual Design of Multistage Launch Vehicles, in: AIAA Guidance, Navigation, and Control Conference and Exhibit, San Francisco, CA, 2005, Paper AIAA 2005-6062.
[21] M. Rosati, A. Gatti, S. Chiesa, A. Teofilatto, L. Mangiacasale, G. Barresi, P. Teofilatto, Launch on demand systems, in: Proceedings of the 20th AIDAA Congress, Milan, Italy, 2009.
[22] W.B. Blake, Missile DATCOM User's Manual—1997 Fortran 90 Revision. United States Air Force, 1998.
[23] A.E. Bryson, Y.C. Ho, Applied Optimal Control, Hemisphere, New York, NY, 1975, pp. 71–89.
[24] D. Hull, Optimal Control Theory for Applications, Springer, New York, NY, 2003, pp. 247–257.
[25] M. Pontani, B.A. Conway, Particle swarm optimization applied to space trajectories, J. Guid. Control Dynam. 33 (5) (2010) 1429–1441.
[26] X. Hu, Y. Shi, R. Eberhart, Recent advances in particle swarm, in: Proceedings of the IEEE Congress on Evolutionary Computation (CEC 2004), Portland, OR, 2004, pp. 90–97.
[27] Z. Michalewicz, M. Schoenauer, Evolutionary algorithms for constrained parameter optimization problems, Evol. Comput. 4 (1) (1996) 1–32.
[28] M. Sirinian, La Traiettoria di lancio del San Marco II. *Atti del Centro Ricerche Aerospaziali*, Università di Roma, Scuola di Ingegneria Aerospaziale, 1967.
[29] LTV Astronautics division, The Scout. PO Box 6267, Dallas, TX, 1963; 〈http://forum.nasaspaceflight.com/index.php?PHPSESSID=ec52e58ff1256bd2adca823c6ebd5c5c&topic=16260.msg377712#msg377712〉.
[30] J.E. Prussing, B.A. Conway, Orbital Mechanics, Oxford University Press, New York, NY, 1993, pp. 46–54.